Dear Editor,

Please consider the manuscript "Silent but deadly? Physics says it depends, memory says mostly" for publication as a Regular Article in *The European Physical Journal Plus*.

The belief that a silent release of flatus smells worse than an audible one is recorded in many languages, yet no mechanistic test of it exists. The closest prior work is a conference abstract on the physics of flatulence (Ancalle et al. 2021). The manuscript follows one passage through a single physical chain, from the valve that makes the sound through dispersion next to the human body to the nose, and asks why a belief arises that the physics does not support. Three results reach beyond the topic.

1. **Routing by the human convective boundary layer.** In simulations of a sofa, a standing pair and a closed lift, the warm air rising along a body takes a slow release next to it and lets a fast one pass, so the momentum of the release, relative to that of the boundary layer, decides which nearby nose it reaches. The same set-up reproduces measured breathing-zone intake from a source at a seated manikin's groin within a factor of 1.5. This bears on any near-body source, such as respiratory, skin and clothing emissions in shared indoor spaces.

2. **A peak bound for linear transport.** We prove that a slower release of the same substance cannot give a higher peak concentration while both are carried passively by the same flow, whatever that flow is. A slower release can win only near the source: through less pre-dilution, by at most the jet-to-ambient velocity ratio, or by being carried to a different nose.

3. **Selection across two senses.** Comparing intensity when both passages are smelled, at driving pressures within the measured physiological range, the answer depends on where the noses are: on a sofa the audible passage is the stronger twice as often as the silent one, while in a closed lift the person standing behind the releaser finds the silent one stronger (55 against 33% of pairs). Yet because a silent passage is noticed only if it is smelled, a remembered silent passage seems the worse one in 91% of comparisons. When an event can be noticed through either of two senses, selection on noticing alone can create a sensory belief stronger than the physics supports, with testable predictions.

The work is applied fluid physics with an interdisciplinary reach into acoustics, sensory science and cognition, the kind of topic *EPJ Plus* welcomes. The outcomes are reported per setting and checked against the scene weights, the observer models, large-eddy peak factors, mesh refinement, release order, clothing and the driving-pressure limit; the online resource reports where the sofa comparison reverses.

Suggested reviewers, with no collaboration with any of them: H. C. Burridge (indoor plume and ventilation modelling, h.burridge@imperial.ac.uk), I. Năstase (human thermal plume and CFD validation, ilinca.nastase@utcb.ro), D. Licina (human convective boundary layer and breathing-zone exposure, author of the manikin data used for validation; dusan.licina@epfl.ch), K. Ito (CFD of inhaled and smelled pollutants near the body, ito@kyudai.jp), S. S. Diwan (turbulent puffs and jets, sdiwan@iisc.ac.in), A. Oleszkiewicz (olfactory perception, anna.oleszkiewicz@uwr.edu.pl) and H. Alves (judgement from selective samples, hans.alves@rub.de).

Proofs and technical details are in Online Resource 1. Code and processed data are openly available at `https://github.com/ccy5123/sbd`. An earlier version of this work is posted as a preprint on engrXiv (`https://doi.org/10.31224/8708`, CC BY 4.0); the manuscript contains no other previously published material and is not under consideration elsewhere. AI-assisted

technologies (Claude Code, Anthropic) were used to write code and simulation set-ups, search the literature and draft text, as documented in the Methods section.

Yours sincerely,

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